Software Engineering

Software Design

M.Sc. Ruben Adrian Gil Rivera

Agenda

- ✓ **Design Definition**
- ✓ **Software Design**
- ✓ **Software Development process**
- ✓ **Design Concepts**
- ✓ **Software Features**
- ✓ **Software Maintenance**

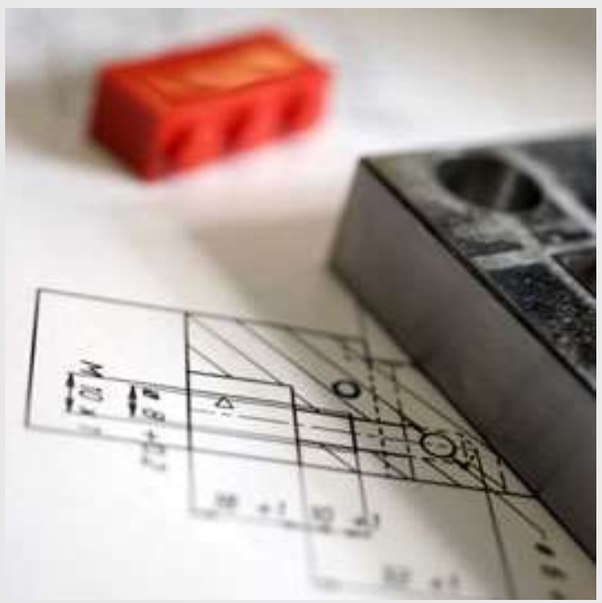


Design

- diseño.

(Del it. disegno).

1. m. Traza o delineación de un edificio o de una figura.
2. m. Proyecto, plan. Diseño urbanístico
3. m. Concepción original de un objeto u obra destinados a la producción en serie. Diseño gráfico, de modas, industrial
4. m. Forma de cada uno de estos objetos. El diseño de esta silla es de inspiración modernista
5. m. Descripción o bosquejo verbal de algo.



Software Design

- Software design is a **process of problem-solving and planning** for a software solution.

- **After** the **purpose and specifications of software are determined**, software developers will design or employ designers to develop a plan for a solution. It **includes** low-level component and algorithm implementation as well as the architectural view.



SW Development process

Software Design

Input

- **Software requirements specification**
- **Change requests** (partly)
- Software verification plan

Output

- **Software architecture description**
  - modules, interfaces, data flow, control flow
  - real-time behaviour, interupt system, OS
- Test specification for integration
- **Module description**
- Test specification for module verification
- Description of private/exported data/operations, algorithms etc.

Activities

- Define software architecture
- Define modules (**note**: there is no distinct detailed design phase)
- Ensure tracebility to software requirements specification
- Use a design language / tools (if available) to avoid ambiguous wording

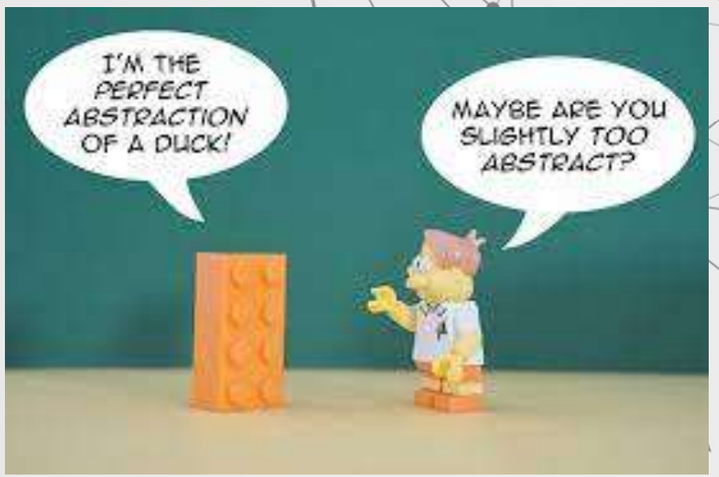


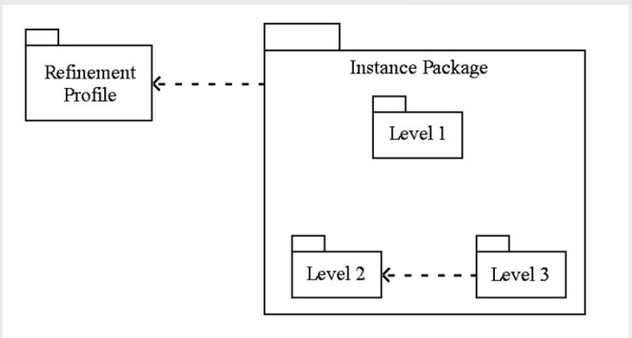
Design Concepts

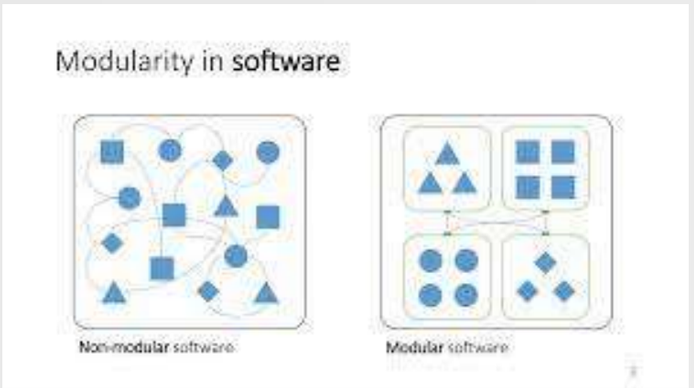
1. **Abstraction** – Abstraction is the process or result of generalization by reducing the information content of a concept typically in order to retain only information which is relevant for a particular purpose.

2. **Refinement** - It is the process of elaboration. **Decomposing** a macroscopic statement of function in a stepwise fashion. Abstraction and Refinement are complementary concepts.

3. **Modularity** - Software architecture is divided into components called modules.







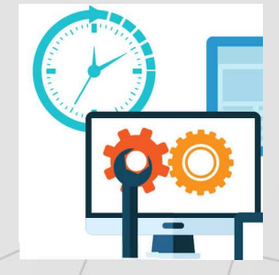
Software Features

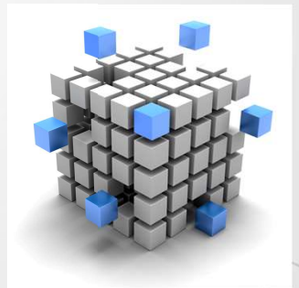




- Ease of reuse: Capability of being reused.
- Portability: Easy to migrate on different Platforms.
- Reliability: How safe it is to use.
- Testability: Capability of being tested
- Efficiency: How good you are at doing a task.
- Flexibility: Being able to combine SW.
- Integrity: Able to protect the system
- Ease of maintenance: SW in optimal conditions.
- Ease of use: Ease of interacting with the system.









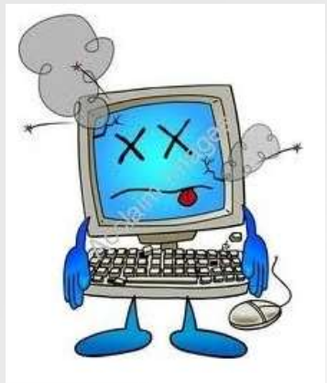




Software Maintenance

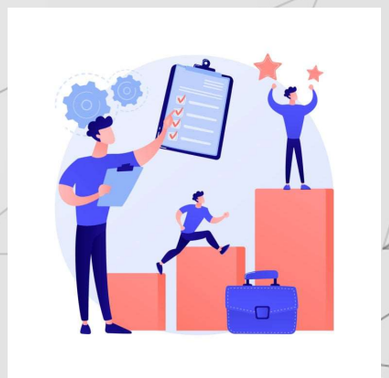
SW maintenance types:

- Corrective
- Adaptive
- Improvement
- Preventive











Questions?

